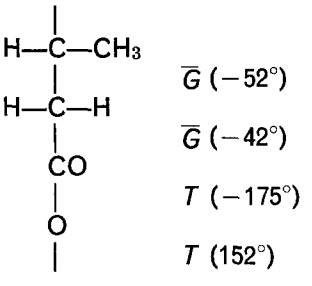
Structural studies of polyesters: 5. Molecular and crystal structures of optically active and racemic poly(β-hydroxybutyrate)

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The molecular and crystal structure of naturally occurring optically active poly(β-hydroxybutyrate) was analysed by X-ray diffraction. The unit cell is orthorhombic, $P2_12_12_1$–D_2^4, with $a=5{\cdot}76$ Å, $b=13{\cdot}20$ Å, and c(fibre period)$=5{\cdot}96$ Å and two molecules pass through the unit cell. The molecule has a conformation shown by the Fischer projection for the case of a rectus polymer, giving a left-handed (2/1) helix.



The crystalline synthesized racemic polymer gave the same X-ray diffraction pattern as the naturally occurring optically active polymer. This result indicates that the racemic polymer has an isotactic configuration and consists of two kinds of crystallites, each composed only of the left-handed helices of rectus polymer chains, or only of the right-handed helices of sinister polymer chains.

INTRODUCTION

Several polyesters of the types

$$[-O(CH_2)_xOCO(CH_2)_yCO-]_n$$

and

$$[-(CH_2)_zCOO-]_n$$

have been studied systematically by the authors using X-ray diffraction, infra-red, and Raman spectroscopic methods ($x=2$ and $y=0$, $x=2$ and $y=2$[1], $x=2$ and $y=4$[2], $z=1$[3], $z=2$[4], and $z=5$[5, 6]). As an extension of these studies, analyses of polyesters with side chains have been undertaken. In the present paper, the structures of optically active and racemic poly(β-hydroxybutyrate) $[-CH(CH_3)CH_2COO-]_n$ are reported. The naturally occurring optically active polymer plays an important role in bacteria as a storage material and is isolated from various bacteria.

There have been several papers on the crystal structures of polyesters with side chains: polypivalolactone $[-CH_2C(CH_3)_2COO-]_n$[7, 8], and optically active poly(β-hydroxybutyrate)[9]. The conformation of the modification (II) of poly(β-propiolactone) $[-(CH_2)_2COO-)_n$ is planar zigzag[4], while the conformation of polypivalolactone is a two-fold helical structure of the type $(G_2T_2)_2$. It would be interesting to compare the molecular conformations of poly(β-propiolactone), polypivalolactone, and poly(β-hydroxybutyrate); these three polymers have the same main chain structure.

In 1967, Okamura and Marchessalt proposed a crystal structure of optically active poly(β-hydroxybutyrate) based upon the qualitatively observed X-ray intensity data[9]. In the present paper, the crystal structures of optically active and racemic poly(β-hydroxybutyrate) are reported in detail. After completion of the present study, the authors were informed through Okamura's review[10] and a personal communication from Marchessalt that these authors had independently found the same crystal structure of the optically active polymer as us.

STRUCTURE DETERMINATION

Samples

Naturally occurring optically active poly(β-hydroxybutyrate) was obtained by extracting *Pseudomonal solanasearum* with chloroform and was purified by reprecipitation with diethyl ether (m.p. 174–177°C; measured under a polarizing microscope). The racemic polymer was prepared by polymerizing DL-β-methyl-β-propiolactone with $[Et_2AlOCPhNPh]_2$ catalyst (m.p. 167–169°C)[11]. The samples for X-ray measurements

were made by heating to just below the melting point and then drawing quickly.

X-ray measurements

Ni-filtered Cu–Kα radiation was used throughout the present X-ray studies. The X-ray photographs of the optically active polymer and the racemic polymer are shown in *Figures 1a* and *b*, respectively. The X-ray diffraction pattern of the racemic polymer is essentially the same as that of the optically active polymer. It was expected that the naturally occurring optically active polymer has a higher regularity than the synthesized racemic polymer. Therefore, for a detailed structure analysis, the photographs of the naturally occurring polymer were used. The meridional reflection data were taken from the photograph using a Weissenberg camera where the sample was oscillated about an axis normal to the fibre axis.

Forty-two independent reflections were observed. The reflection intensities of the fibre diagram obtained by the multiple film method were measured by visual comparison with a standard intensity scale. The corrections for Lorentz polarization factors and the obliquity effect in the fibre diagram were made.

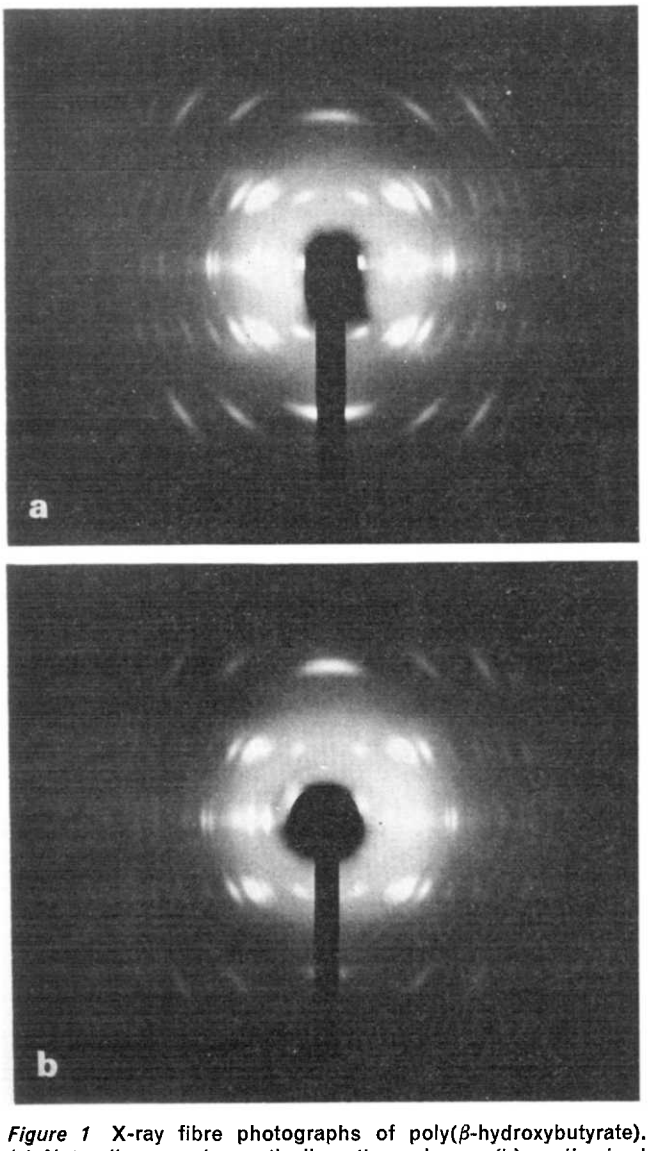


Figure 1 X-ray fibre photographs of poly(β-hydroxybutyrate). (a) Naturally occurring optically active polymer; (b) synthesized racemic polymer

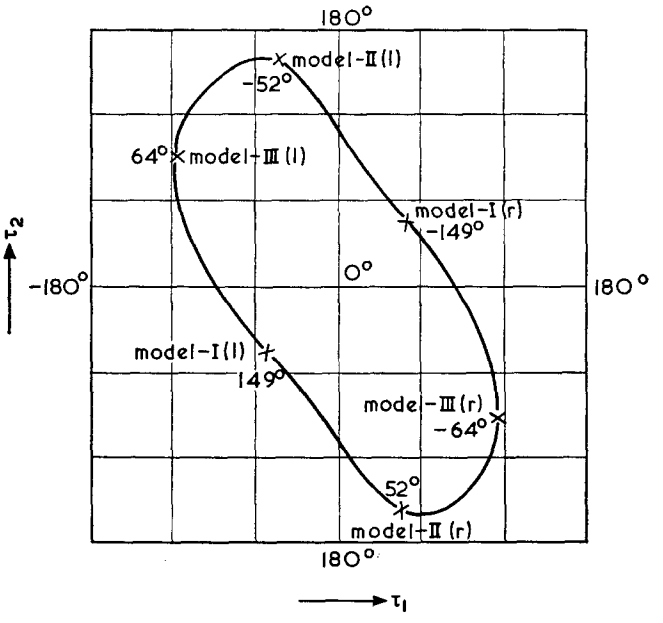


Figure 2 Three-dimensional closed curve showing the relationship between τ_1, τ_2, and τ_4 for possible conformation of the main chain of poly(β-hydroxybutyrate)

Unit cell and space group

The reflections observed on the photograph could be indexed by an orthorhombic cell; $a = 5{\cdot}76$ Å, $b = 13{\cdot}20$ Å, and c (fibre period) $= 5{\cdot}96$ Å. These values were calibrated by using the spacings of the reflections of aluminium powder and they agreed with those reported by Okamura and Marchessalt[9]. It was assumed that there are four monomeric units in a unit cell, by comparing the calculated density of 1·26 g/ml with the observed density of 1·23–1·25 g/ml measured by the flotation method[12]. The following systematic absences were found; $h00$ when h is odd, $0k0$ when k is odd, and $00l$ when l is odd. Therefore the possible space group is $P2_12_12_1$–D_2^4. This space group is a suitable one for an optically active material. It was assumed from these experimental data that this polymer molecule has a two-fold screw axis relating two monomeric units, and two molecules pass through the unit cell.

Molecular models

Here the rectus polymer model was assumed, since the naturally occurring optically active poly(β-hydroxybutyrate) is considered to be rectus because hydrolysis and autolysis of this polymer give D(−)-β-hydroxybutyric acid[13]. At first, the molecular models with two-fold screw axis were constructed by assuming the following bond lengths and bond angles: C_I–$C_{II} = C_I$–$C_{IV} = 1{\cdot}54$ Å, C_{II}–$C_{III} = 1{\cdot}52$ Å, C_{III}–$O_I = 1{\cdot}36$ Å, C_{III}–$O_{II} = 1{\cdot}23$ Å, O_I–$C_I(+1) = 1{\cdot}43$ Å, $\angle O_I(-1)C_IC_{II} = \angle O_I(-1)C_IC_{IV} = \angle C_{IV}C_IC_{II} = \angle C_IC_{II}C_{III} = 109{\cdot}5°$, $\angle C_{II}C_{III}O_I = 115{\cdot}0°$, $\angle C_{II}C_{III}O_{II} = \angle O_{II}C_{III}O_I = 122{\cdot}5°$, and $\angle C_{III}O_IC_I(+1) = 112{\cdot}0°$. The numbering of the atoms and the internal rotation angles are as follows;

$$-O_I(-1)-\overset{\tau_1}{C_I}-\overset{\tau_2}{C_{II}}-\overset{\tau_3}{C_{III}}-\overset{\tau_4}{O_I}-C_I(+1)-$$
$$\qquad\qquad\quad |\qquad\quad ||$$
$$\qquad\qquad C_{IV}\qquad O_{II}$$

By assuming the two-fold screw symmetry, the number of independent internal rotation angles is three. From the results of the structure analyses of polyesters of the types

$[-O(CH_2)_xOCO(CH_2)_yCO-]_n$ and $[-(CH_2)_zCOO-]_n$, it is reasonable to fix τ_3 at 180·0°[1]. By fixing τ_3 at 180·0° and varying τ_1 and τ_2 from −180° to 180·0°, the values of τ_4 for all possible molecular models were obtained which satisfy the identity period of 5·96 Å. A set of the internal rotation angles was on one closed curve in a cube defined by the three-dimensional Cartesian coordinates, τ_1, τ_2, and τ_4, each covering from −180·0° to 180·0°. In *Figure 2*, the projection to the $\tau_1\tau_2$ plane is shown and τ_4 values of several points are given on the closed curve. From this Figure, six models with plausible internal rotation angles (nearly *trans* (*T*), skew (*S*), and *gauche* (*G*)) could be obtained: three left-handed helices and three right-handed helices. The internal rotation angles of the left-handed helices are given as follows and the right-handed helical models [model I(r), model II(r), and model III(r)] can be given by changing the sign of all the internal rotation angles of the left-handed helical models;

Model	τ_1(°)	τ_2(°)	τ_3(°)	τ_4(°)
I(*l*)	−50	−45	180	149
II(*l*)	−45	159	180	−52
III(*l*)	−115	92	180	64

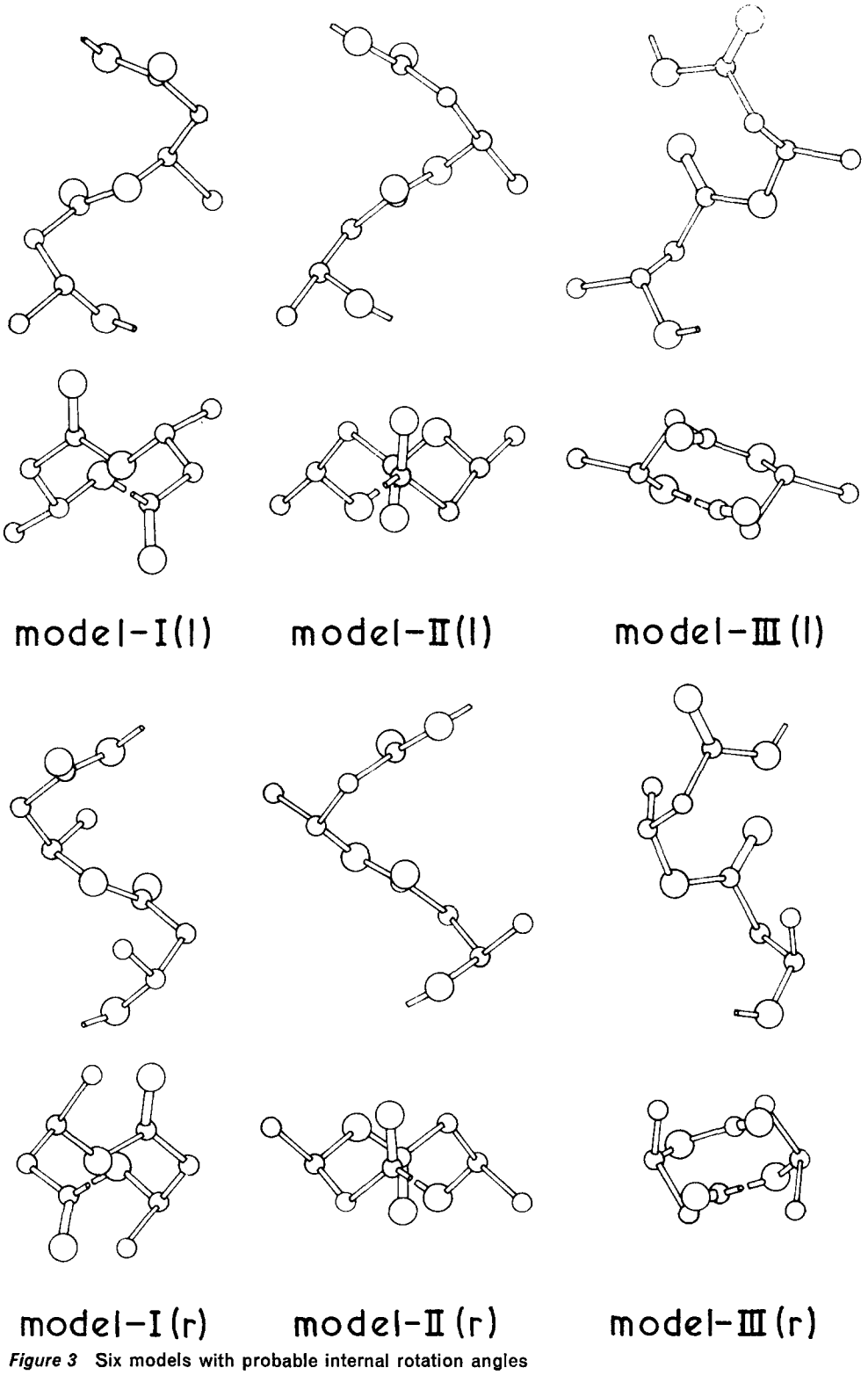


Figure 3 Six models with probable internal rotation angles

These six rectus polymer models are shown in *Figure 3*.

Structure factor calculations

At first, the structure factors of the equatorial reflections for the six models were calculated. Good agreement between observed and calculated structure factors could be obtained only for the structure in which model I(l) was located so that the direction connecting two methylene groups on the *ab* projection is nearly parallel to the *b* axis. The model I(l) is considered the most probable molecular conformation. But, there remains another parameter, the translation along the *c* axis (parallel to the fibre axis). Referring to the observed intensities of the meridional reflections, the model I(l) was placed as the *z*-coordinate of the carbonyl carbon atom to be nearly zero. At this stage, the contribution of the hydrogen atoms was included in the structure factor calculations, by assuming the C–H bond length to be 1·09 Å and the ∠CCH and ∠HCH bond angles to be 109·5°. As a result, the calculated structure factors for the *all*

Table 1 Atomic coordinates and thermal coefficients

	x	*y*	*z*	*B*(Å²)
C_I	0·437	−0·068	0·217	4·7
C_{II}	0·318	−0·139	0·389	3·3
C_{III}	0·143	−0·077	0·513	5·1
C_{IV}	0·587	−0·140	0·052	7·1
O_I	0·232	0·017	0·585	5·1
O_{II}	−0·057	−0·105	0·558	5·7
Assumed values for hydrogen atoms				
H(I)	0·557	−0·020	0·294	10·0
H(II-1)	0·447	−0·171	0·499	10·0
H(II-2)	0·226	−0·200	0·301	10·0
H(IV-1)	0·667	−0·092	−0·076	12·0
H(IV-2)	0·464	−0·192	−0·036	12·0
H(IV-3)	0·710	−0·182	0·143	12·0

Table 2 Comparison of obs. and calc. structure factors for naturally occurring optically active poly(β-hydroxybutyrate)

h	*k*	*l*	$(I_o)^{1/2a}$	$(I_c)^{1/2b}$
0	2	0	86·3	70·0
1	1	0	85·5	86·3
1	2	0	(5·1)[c]	17·3
1	3	0	38·6	43·3
0	4	0	50·9	49·6
2	0	0	28·3	25·6
1	4	0		
2	1	0		
2	2	0	21·8	18·2
2	3	0	38·3	28·2
1	5	0		
0	6	0	34·7	42·3
2	4	0		
1	6	0	21·6	12·1
2	5	0	(8·4)	2·0
3	1	0	29·1	35·6
3	2	0	(8·7)	14·6
1	7	0	(8·8)	9·8
3	3	0	39·0	36·5
2	6	0		
3	4	0	26·1	18·2
0	8	0		
1	8	0	27·3	28·6
2	7	0		
3	5	0	(9·9)	5·0
3	6	0	25·6	23·1
4	0	0		
2	8	0		
4	1	0		
1	9	0		
4	2	0		
4	3	0	(10·8)	14·0
3	7	0	29·4	28·9
0	10	0		
0	1	1	24·3	23·3
0	2	1	43·3	36·8
1	0	1	54·7	53·9
1	1	1	62·9	79·1
0	3	1	55·6	59·8
1	2	1		
1	3	1	38·8	33·3
0	4	1	10·5	9·3
2	0	1	38·8	36·7
1	4	1		
2	1	1		
0	5	1	31·5	24·9
2	2	1		
2	3	1	47·3	42·4
1	5	1		
0	6	1	46·3	41·0
2	4	1		
1	6	1	26·0	26·0
2	5	1	22·2	18·1
3	0	1		
3	1	1	30·8	28·9
0	7	1		
3	2	1	(4·6)	4·4
1	7	1	(4·7)	8·8
3	3	1	23·9	25·4
2	6	1		
3	4	1	23·8	13·8
0	8	1		
1	8	1	(5·2)	4·3
2	7	1	(5·2)	3·1
3	5	1	(5·3)	10·6
0	9	1	26·3	34·3
3	6	1		
4	0	1		
2	8	1		
4	1	1		
1	9	1		
0	3	2	47·6	57·3
1	2	2		
1	3	2	27·7	26·2
0	4	2	28·7	22·5
2	0	2	63·9	66·7
1	4	2		
2	1	2		
0	5	2	25·4	23·9
2	2	2		
2	3	2	49·7	44·1
1	5	2		
0	6	2	28·7	26·9
2	4	2		
1	6	2	(8·5)	6·7
2	5	2	41·6	46·8
3	0	2		
3	1	2		
0	7	2	(9·0)	1·8
3	2	2	(9·2)	15·2
1	7	2	(9·3)	13·3
3	3	2	19·7	11·1
2	6	2		
3	4	2	(9·9)	13·3
0	8	2		
1	8	2	32·8	22·2
2	7	2		
3	5	2		

[a] The observed structure factors $(I_o)^{1/2}$ were put on the same scale as the $(I_c)^{1/2}[=(mF_c^2)^{1/2}]$ values by setting $\Sigma k(I_o)^{1/2}=\Sigma(mF_c^2)^{1/2}$ where k is the scale factor and m is the multiplicity

[b] $(I_c)^{1/2}$ values of the reflections which overlap on the X-ray fibre photograph are $(\Sigma mF_c^2)^{1/2}$

[c] For non-observed reflections, $(I_o)^{1/2}$ values were assumed to be half of the most weakly observed intensity

reflections satisfactorily agreed with the observed structure factors. The discrepancy factor R

$$= \Sigma|(I_o)^{1/2} - (I_c)^{1/2}| \, \Sigma(I_o)^{1/2}$$

was 15·6% (only for forty-two observed reflections). Then, the atomic coordinates and thermal coefficients were refined by the diagonal least-squares method. The discrepancy factor R was reduced to 13·5%. *Table 1* lists the atomic coordinates and thermal coefficients. The observed and calculated structure factors are given in *Table 2*.

For all computations in the present study, the NEAC-2200 electronic computer in this university was used.

RESULTS AND DISCUSSION

Optically active poly(β-hydroxybutyrate)

The molecular conformation of naturally occurring optically active poly(β-hydroxybutyrate) was found to be approximately $(\bar{G}_2T_2)_2$. In *Figure 4* the molecular dimensions of this polymer are shown. The values of the bond lengths and bond angles are reasonable. The crystal structure of this optically active polymer is shown in *Figure 5*. In the case of polypivalolactone, right- and left-handed molecules coexist in the crystal lattice. On the other hand, the crystal lattice of the optically active poly(β-hydroxybutyrate) contains two left-handed helical molecules in antiparallel orientation. The oxygen atoms of the carbonyl groups are located nearly at the same level, $z=0$ and 1/2. No abnormal atomic distances were found and the packing of the molecules was reasonable.

Racemic poly(β-hydroxybutyrate)

It has already been mentioned that the X-ray photograph of the synthesized crystalline racemic poly(β-hydroxybutyrate) was essentially the same as the naturally occurring optically active polymer (*Figures 1a* and *b*). Accordingly, it may be concluded that the crystalline polymer prepared from the racemic monomer is isotactic. Naturally occurring optically active polymer was found to be a left-handed helix because the configuration of the β-carbon atom is rectus, and the sinister polymer should be right-handed. From these results based on the X-ray analysis, it is conceivable that the bulk sample of this racemic polymer consists of two kinds of crystallites, each composed only of the rectus polymer chains, or only of the sinister polymer chains. Such a type of polymer has already been reported by the authors in the case of poly(propylene sulphide)[14].

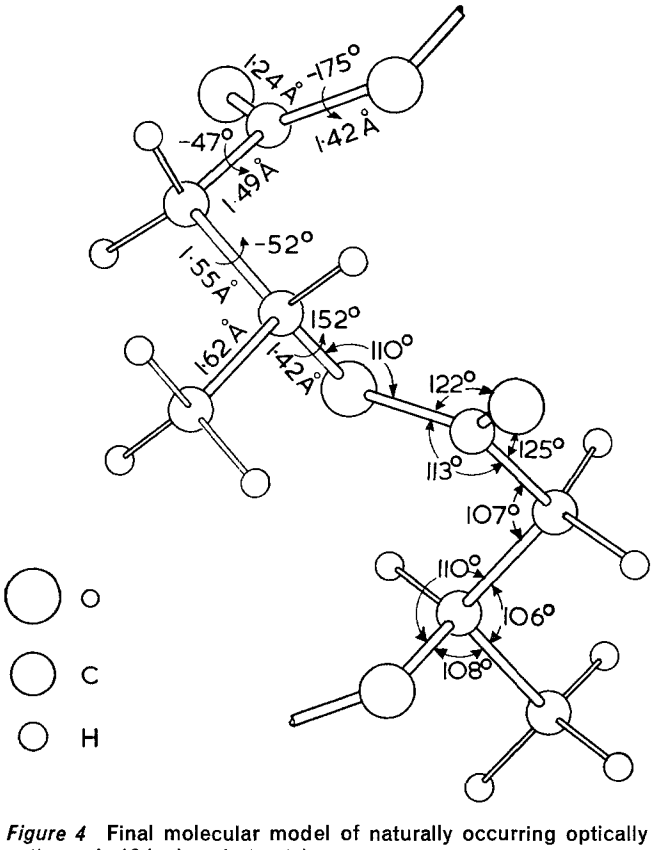


Figure 4 Final molecular model of naturally occurring optically active poly (β-hydroxybutyrate)

Molecular conformations of polyesters

The skeletal conformations of some polyesters analysed so far are summarized below:

	—C—C—O—C—C—C—
Poly(β-propiolactone)[4]	*T T T T T*
Poly(ε-caprolactone)[5]	*T T T T T*
Poly(ethylene adipate)[15]	*T S T T T*
Poly(ethylene suberate)[15]	*T S T T T*
Poly(ethylene succinate)[1]	$\bar{G}$ *T T T G*
Poly(tetramethylene succinate)[2]	*G T T T T*
Polypivalolactone[7,8]	*G T T G G*
Poly(β-hydroxybutyrate)	*G T T G G*

	—C—C—O—C—C—O—
Poly(ethylene oxalate)[1]	*G T T T T*

The C(O)–O bond of all these polyesters has *trans* conformation. In the cases of the polyesters of the types $[-O(CH_2)_xOCO(CH_2)_yCO-]_n$ and $[-(CH_2)_2COO-]_n$, the C–C(O) bond is stable in the *trans* form and also O–C bond preferably takes the *trans* form. In the cases of polypivalolactone and poly(β-hydroxybutyrate), which have side chains, the C–C(O) bond is in the *gauche* form. But the C(O)–O and O–C bonds are in the *trans* form just as in the cases of the polyesters with no side chains. This suggests that the conformation of C–C(O) bond in a polyester is liable to be affected by introduction of substituent groups. Poly(β-hydroxybutyrate) and polypivalolactone take $(G_2T_2)_2$ conformation. Such a $(G_2T_2)_2$ type conformation was also found in the cases of the modification (III) of polyoxacyclobutane[16] and syndiotactic polypropylene[17].

Disordered modification

When a sample of poly(β-hydroxybutyrate) was stretched even further after necking, there appeared another modification with a fibre period of 4·70 Å. *Figure 6* is a fibre photograph of a sample containing this modification, which shows a sharp reflection on the equator and streaks on the layer lines. This feature may suggest this modification is in a kind of paracrystalline structure in which the molecular chains are arranged regularly in the lateral direction, but irregularly along the fibre axis. From the observed fibre identity period (4·70 Å) it is considered that this modification has a twisted planar zigzag conformation. A detailed structure analysis is now being made.

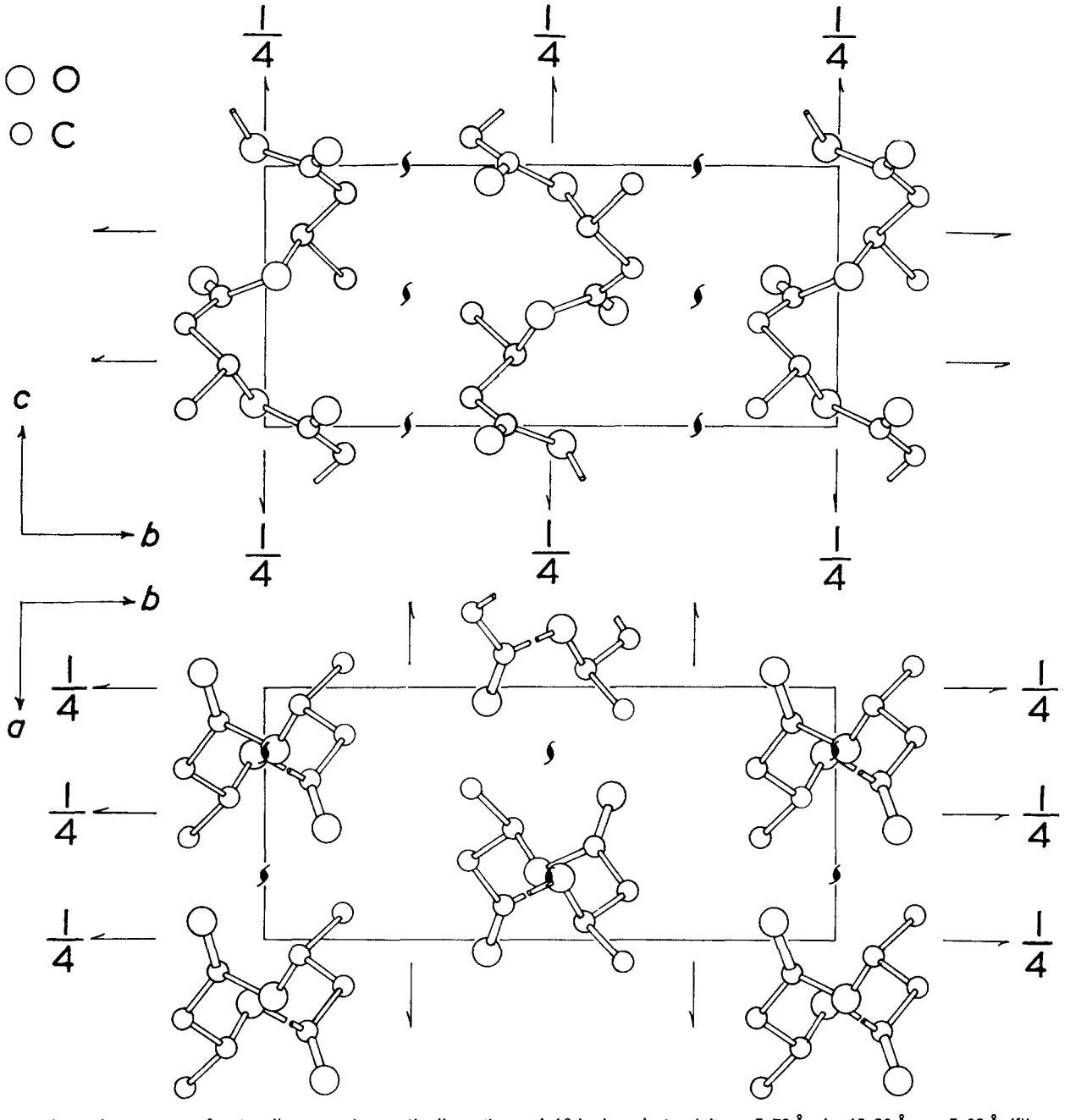


Figure 5 Crystal structure of naturally occurring optically active poly(β-hydroxybutyrate). a=5·76 Å; b=13·20 Å; c=5·96 Å (fibre axis)

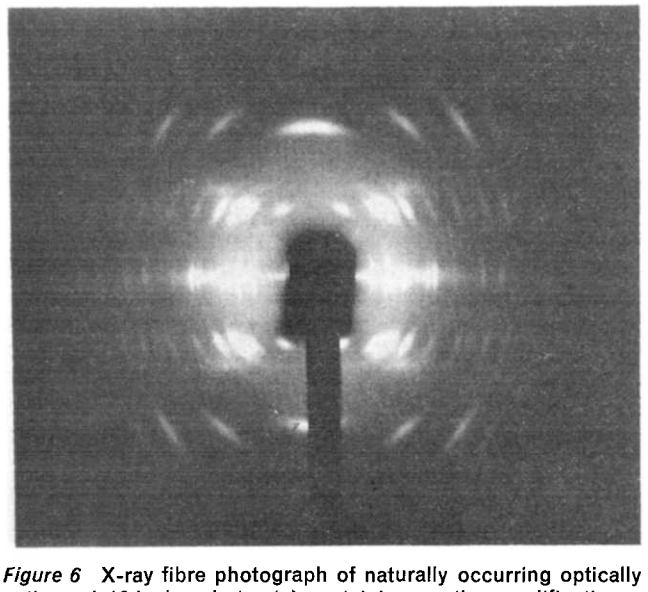


Figure 6 X-ray fibre photograph of naturally occurring optically active poly(β-hydroxybutyrate) containing another modification

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